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REGULAR ARTICLE

The Relationship Between School Organizational Characteristics and Reliance on Out-of-Field Teachers in Mathematics and Science: Cross-National Evidence from TALIS 2008

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Abstract This article investigates the effect of schools' organizational features on teacher staffing decisions at the secondary level cross-nationally. I examine the importance of the principal's administrative leadership style and school-level autonomy in making staffing decisions to understand variation in assignments of out-of-field teachers (OFT) to teach mathematics and science. I conduct secondary analysis based on a new international data set compiled by the Organisation for Economic Co-operation and Development on lower-secondary school administration and teachers in 15 countries. I utilize a country fixed-effects model to study the effect of different types of organizational characteristics on the extent of reliance on out-of-field teaching in mathematics and science cross-nationally. The results indicate that the use of OFT is a school-specific issue. Several school characteristics affect levels of out-of-field teaching. After controlling for country fixed effects, I do not find a systematic relationship between the principal's administrative leadership and the school's reliance on OFT. However, higher school-level autonomy in making staffing decisions is significantly associated with lower reliance on out-of-field teaching.

Keywords International comparison · Out-of-field teaching · Teacher assignment · Institutional arrangements

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Introduction

The idea that out-of-field teachers (OFT) are underqualified is not new to educators. This label refers to teachers who teach a subject that lies outside their formal academic training. In the United States, educators are aware that out-of-field teaching is deeply rooted in schools (Hechinger 1985). OFTs are commonly deployed in a wide range of subjects (Ingersoll 1999). Researchers report that out-of-field teaching exists in many other countries as well [Organisation for Economic Co-operation and Development (OECD) 2005; UNESCO Institute for Statistics 2006]. Out-of-field teaching is considered problematic mainly because of increasing requirements for teachers to master higher levels of subject matter knowledge. From a teacher's knowledge perspective (Shulman 1986), OFTs are seen as deficient in this particular dimension. Although OFTs may still have a good command of general pedagogy, they are by definition lacking in subject matter training. Thus, it is very hard to argue that OFTs should develop the full ability to teach their subjects without any additional assistance. In this sense, the phenomenon of OFTs poses a challenge to the efforts that have been made to improve teacher quality, particularly teacher competency in the subjects taught.

Conventional views typically treat out-of-field teaching as a product of low teacher quality or teacher shortage. More recent studies in the United States have indicated that out-of-field teaching could be a school management issue pertinent to particular school characteristics (Ingersoll 1999, 2005, 2006). However, we know little about the importance of particular school factors to explaining the variation in out-of-field teaching between schools. In addition, because most empirical studies of out-of-field teaching are conducted in the US context, it is not clear

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whether the findings could be generalized to other countries. In this paper, I use data from a comprehensive international teacher survey conducted by OECD to study the importance of organizational features in determining the extent of out-of-field teaching in schools. By using an international comparative data set, I exploit variations in school organizational factors to gauge their effects on out-of-field teaching assignments in math and science. Benefiting from this rich data set, I was able to control for important variables that have been identified in teacher labor market research as crucial to staffing. I hope to contribute to the growing body of knowledge on teacher quality and school management.

Literature Review

Prevalence of Out-of-Field Teaching

In the United States, where statistics on out-of-field teaching have been compiled for the past two decades, the National Center for Educational Statistics (NCES) estimates that out-of-field teaching has been a widespread and persistent issue since the late 1980s. At the high school level, out-of-field teaching has remained steady at about 20–25 % of mathematics instruction from the 1990s to 2008, the most recent year for which statistics are available. In science subjects overall, the estimates range from 13 to 20 % during the same period of time. Within individual subject areas in science, the situation is much worse. In physics, 42–67 % of teachers are out-of-field. In chemistry, about 50–65 % of teachers are teaching out-of-field. In biology/life science, the estimates are 24–42 % (Morton et al. 2008; Schools and Staffing Survey 2010; Seastrom et al. 2002). These numbers show that a large proportion of math and science teachers did not obtain proper training in the subject matter fields. The phenomenon is less of an issue in the humanities and social sciences. About 15–23 % of English teachers are out-of-field, and about 16–24 % of social sciences teachers are out-of-field. While statistics on OFTs at the high school level are most readily available, Ingersoll (2003) suggests that out-of-field teaching is even more prevalent in the middle school grades (grades 7 and 8) because some states require these teachers to hold an elementary school-type certificate that emphasizes broad teaching skills.

Outside the United States, a 2006 UNESCO report showed that a large proportion of students in countries such as Norway and Canada were taught in mathematics by OFTs (UNESCO Institute for Statistics 2006). An OECD report pointed out that lowering subject matter qualification requirements is often used as a means to cope with teacher shortages in certain areas (OECD 2005). In countries such as the Slovak Republic, it is estimated that "25 % of primary classes, 30 % of lower secondary classes, and 15 % of vocational classes were taught by teachers who did not have teaching qualifications, or were teaching out-of-field, or who had already reached official retirement age" (OECD 2005, p. 49). Statistics from the Trends in International Math and Science Study (TIMSS) offer another peek into out-of-field teaching practice worldwide. They show that out-of-field teaching in math and science is an international phenomenon that is widespread in many educational systems (Akiba and LeTendre 2009; Akiba et al. 2007).[1]

Factors Explaining Out-of-Field Teaching

Quality of Teacher Supply

Several arguments have emerged in the debate on the origins of widespread out-of-field teaching assignments, all in the US context. The first argument arose from a report by the National Commission on Teaching and America's Future (NCTAF) in (1996). The report claimed that out-of-field teaching was a result of inadequate teacher training (NCTAF 1996). Commentators argued that teacher education in the United States is not rigorous enough to equip future teachers with sufficient knowledge for teaching. The implication is that because teacher education in general is of poor quality, it makes little sense for the employers (i.e., districts and schools) to differentiate among candidates with different subject matter backgrounds, thus leading to widespread out-of-field assignments. Such a view echoes the rhetoric of declining teaching quality in the United States. But it fails to address why out-of-field teaching continues to thrive in an era of increasing teacher qualification. Indeed, statistics show that most teachers in the United States have at least a bachelor's degree, and a growing number hold a master's degree (Ingersoll 2003). The number of teachers with no certification or license is falling (Ingersoll 1999). Even after the passage of strong subject matter qualification requirements in the No Child Left Behind Act of 2002, levels of out-of-field teaching did not seem to change dramatically (Schools and Staffing Survey 2010).

[1] I have to remind readers that because of the way in which the TIMSS teacher survey is constructed, the teacher sample is not nationally representative, but the student sample is. Any generalized reference to teacher characteristics must be made in association with the student population, which can be quite cumbersome in the out-of-field teaching case.

Quantity of Teacher Supply

According to this view, the deployment of OFTs is due to shortages of teachers in certain subject fields. Schools are forced to assign teachers from non-shortage disciplinary areas to teach in those areas where teacher supply cannot meet classroom demand. In fact, teacher shortage is widely believed to be the main driver of out-of-field assignments in a number of countries (Eurydice 2002; OECD 2005). However, there are conflicting findings. Ingersoll (1999) found that some schools with no difficulty in staffing nonetheless reported similarly high levels of out-of-field teaching. In addition, out-of-field teaching existed in subjects that traditionally have a surplus of teachers, such as English and social studies (Ingersoll 1999). More recent studies also reveal the complicated nature of teacher shortage. Ingersoll and Perda (2010) argued that teacher shortage is not the result of a lack of candidates in the teacher preparation pipeline; rather, it is the high turnover rate in certain types of schools and subject areas that poses a challenge to schools and districts making staffing decisions. Consequently, out-of-field assignments may reflect areas of higher teacher turnover, not directly teacher undersupply.[2] While in theory teacher shortages can be one of the reasons for out-of-field teaching, the link between these two phenomena has not been formally tested.

The Organizational Perspective

Since the previous two arguments fall substantially short of explaining the out-of-field teaching phenomena, a third explanation has emerged. In a series of studies, Ingersoll (1999, 2002, 2003, 2005, 2006) argued that out-of-field teaching should be understood as an organizational issue. Out-of-field assignment is not due solely to emergencies, but rather is a common practice in school organization. In a country such as the United States, school administration is fundamentally decentralized. Decisions about staffing and hiring are typically made within a school or a district, but the reasons for these decisions are rarely scrutinized directly. In such a situation, the school principal has broad discretionary powers in terms of assigning teachers to teaching posts. By contrast, teachers have little authority to determine their own positions. In many cases, teaching assignments are based on convenience or economic concerns. To match each teacher's expertise with specific student learning needs is a time-consuming endeavor and principals, in many situations, have little incentive to undertake it because their compensation and evaluation is not affected by the results of these decisions. Burdened by other issues, principals may make decisions based on their perception of a teacher's ability to teach certain subjects, and in some cases the decision may be based simply on convenience. Although subject matter competency is an important factor in hiring decisions for many administrators, there is no guarantee that once these teachers are hired, they will be properly assigned to the teaching posts that need them the most. This third hypothesis predicts that a principal's role is important in explaining variation in out-of-field teaching.

To summarize, while the exact statistic is limited on out-of-field teaching globally, there is emerging evidence suggesting that OFTs are deployed in many nations. Past studies have suggested that certain schools experience high reliance on out-of-field teaching. Although the school perspective leads researchers to examine particular features of school organization, there is no consensus on the precise organizational characteristics that might affect the deployment of OFTs. Most research on the distribution of OFTs is largely conducted under the US setting. Whether such findings could be generalized to other nations remains to be verified. This gap in research is largely due to lack of comparable cross-national data. Methodologically, since school characteristics could be determined endogenously by institutional settings within a country, previous studies conducted in a single country setting, where schools are more likely to be similar to one another, cannot easily isolate the effect of institutional characteristics on out-of-field teaching.

In light of the gap existing in current literature, I attempt to answer two research questions:

1. What is the level of out-of-field teaching in math and science internationally?
2. What are the organizational characteristics of schools with high levels of out-of-field assignments in math and science?

The implication of the answer to this question will deepen our understanding on school management and teacher quality. By pinpointing the particular characteristics of schools that deploy high number of OFTs, policy makers and researchers could move forward to design targeting measure to improve the staffing situation there.

Data and Methods

Data Source

I rely on data from OECD's Teaching and Learning International Survey 2008 (TALIS 2008). TALIS is the

[2] In economics, any imbalance between labor supply and demand can be labeled a shortage. However, in conventional usage, shortage typically refers to insufficient supply. Ingersoll and Perda (2010) made this important distinction in their discussion. Though not stated explicitly, previous discussions on out-of-field teaching typically refer to teacher shortage as undersupply of teachers, which may not be the sole explanation for vacancies in certain teaching positions.

"first international survey to focus on the working conditions of teachers and learning environment in schools" (OECD 2009, p. 18). TALIS includes lower secondary school and teacher-level data from 24 countries, more than 4,000 schools, and more than 70,000 teachers. The data is collected during October–December 2007 for Southern Hemisphere countries and March–May 2008 for Northern Hemisphere countries. In my study, I use a subsample of 15 countries.[3] The teachers surveyed in TALIS all teach at the lower secondary level.[4] The TALIS teacher sample excludes special education, substitute, emergency, and occasional teachers as well as those who took long-term leave (OECD 2010). Typically, 20 teachers were sampled per school and 200 schools were sample per country. The TALIS questionnaires are filled either online or on paper depending on country's choice. According to OECD (2009), the response rate is 75 % from all sampled teachers in the country.

The final samples are described in Table 1. In this analysis, I focus on public school teachers in math and science subjects. I choose to focus on math and science teachers because their supply and demand structure is likely to be different from that in other disciplines. The demand for workers who have a background in math and science is particularly high in service and high-end manufacturing sectors. Such occupations outside of teaching that demand a great amount of math and science skills typically offer higher wages. The wage differential thus causes math and science teaching to be more susceptible to teacher shortages than the teaching of other subjects (Ladd 2007).

Data Measures

Dependent Variable

In the teacher questionnaire, teachers were asked whether their current subjects were part of their academic training (Question 36). This was a yes/no type of question. I used the answer to this question as the out-of-field teaching indicator. Using this variable, I calculated the percentage of OFTs within the total population of teachers in math and science, respectively, by school. This measure, calculated using Eq. 1, became my dependent variable in the subsequent analysis.[5]

$$\text{Percent OFT per school} = \frac{\text{Total number of math and science OFT per school}}{\text{Total number of math and science teachers per school}}. \quad (1)$$

Independent Variable

Three independent variables are key to my analysis: the principal's administrative leadership, autonomy in hiring and determining teacher salaries, and personnel shortage. All three variables are measured on a continuous scale at the school level.

I use the administrative leadership variable as a measure of the principal's role in school management. In the TALIS database, the administrative leadership variable measures two dimensions of management: accountability management and bureaucratic management. The TALIS team first constructed a continuous index for each of the two dimensions to transform the ordinal items into two continuous variables using confirmatory factor analysis (CFA). Then they took the simple average of the two component management indices and normalized the composite score across countries to obtain an international metric with a mean of zero and a standard deviation of one (OECD 2010). All of the indices have high reliability and reasonable model fit. The administrative leadership variable "relate[s] to school administrative tasks, enforcing rules and procedures, and accountability role of the school principal" (OECD 2010, p. 146).

I include the autonomy-of-hiring variable to measure the level of autonomy each school has in making staffing decisions. The autonomy-of-hiring variable contains five elements as described in OECD (2010). Similarly, the shortage-of-personnel variable measures whether a school lacks various types of working personnel. The autonomy-of-hiring variable and shortage of school personnel variable are constructed separately using principal component analysis. These two variables are also normalized across countries with an average of zero and a standard deviation of one. For the autonomy variable, higher values on the variable indicate a higher level of school responsibility in

[3] There are 24 countries in the original TALIS sample. I exclude nine countries for various reasons. Iceland withdrew from the TALIS database after the data were released due to quality concerns. Italy was excluded because the teacher-level subject matter identifier variable was completely missing. Australia, Austria, Denmark, Ireland, Malaysia, the Netherlands, and Poland are excluded because their first stage sampling unit (school) was not chosen based on the number of teachers, which could potentially affect the generalizability of the findings.

[4] All teachers surveyed are ISCED 2 level teachers. An ISCED 2 teacher may teach in a range of grades depending on the national context.

[5] There are certainly limitations to this measure. For instance, even though it was common for teachers to take multiple assignments at lower secondary levels, the teacher questionnaire does not differentiate between primary and secondary teaching assignments. Thus, I was not able to discern whether OFT are teaching multiple subjects at the same time. I was also not able to differentiate out-of-field teaching by assignment type. The school-level estimates use a generated school-level sampling weights of teachers by separating the school weight variable (SCHWGT) from the teacher weight variable (TCHWGT).

Table 1 Description of TALIS 2008 sample

Country	Grades sampled	Total schools sampled	Total teachers sampled	Public schools sampled	Math and science teachers sampled
Brazil	5–8 or 6/9	380	5,834	257	1,653
Bulgaria	5–8	199	3,796	193	1,036
Estonia	7–9	195	3,154	189	730
Hungary	5–8	183	2,934	163	728
Korea	7–9	171	2,970	110	738
Lithuania	5–10	206	3,535	200	761
Malta	Forms I–V	58	1,142	30	265
Mexico	7–9	192	3,368	149	930
Norway	8–10	156	2,458	145	618
Portugal	ISCED 2[a]	173	3,046	130	857
Slovak Rep	5–9	186	3,157	158	872
Slovenia	7–9	184	3,069	177	843
Spain	1–4 compulsory secondary	193	3,362	138	881
Turkey	6–8	193	3,224	140	707
Belgium (F)	First stage of secondary education	197	3,473	83	667

[a] ISCED 2 in Portugal consists of the last 3 years of compulsory education

hiring and determining teacher salaries. For the shortage variable, higher values indicate relatively higher inadequacy of school personnel.

Other school-level variables that I used in the analysis include the location of the school, school size, the average level of parental education as a proxy of student's social economical background, the percentage of female teachers, the percentage of part-time teachers, the percentage of teachers with short-term contracts, teachers' educational attainment, and amount of teaching experience. I include these variables in the statistical analysis based on previous studies that indicate their relevance. The variables are summarized in Table 2.

Data Analysis

I specify two sets of models. The first set of models consists of linear models based on standard ordinary least square (OLS) assumptions. Regression models based on Eqs. 2 to 11 are considered by pooling all of the school observations across countries. The outcome variable OFT_{is} measures the percentage of teachers who teach out-of-field in math and science in school i in country s. I control for country fixed effects a_s using dummy variables. This approach allows me to account for all of the variation in school-level out-of-field teaching due to country characteristics. It is commonly used in the international education literature (Ammermueller and Pischke 2009; Chudgar et al. 2013; Hanushek and Woessmann 2010; Ohinata and van Ours 2011). My primary interests are the coefficients on three policy variables, *ADMINL*, *AUTHIRE*, and *LACKPERS*, which stand for the principal's administrative leadership, the school's autonomy in hiring teachers and determining their salaries, and the degree to which a school lacks personnel, respectively. I started fitting models using one of these three variables at a time as described in Eqs. 2 to 4, in which I also include school controls as described in Table 3. I fitted two models for each variable of interest. The first model is fitted without country fixed effects and the second model is fitted with country dummy variables. The rationale of using country fixed effects is that institutional arrangements are most likely endogenous to each country. That is, they are correlated with a set of cultural and sociopolitical rules that are determined within each nation and not explicitly available to researchers. The use of country fixed effects can thus control for the variation in outcome variables that is caused by these unobserved, country-specific factors. By comparing the estimates from models with and without fixed effects, we can gauge the extent to which the national context affects a school's decision to use OFTs. I repeat this strategy in a full specification by using all three variables in one model.

$$\mathrm{OFT}_{is} = \delta + \alpha\mathrm{ADMINL}_{is} + \gamma\mathrm{SCHOOL}_{is} + a_s + \varepsilon_{is} \quad (2)$$

$$\mathrm{OFT}_{is} = \delta + \alpha\mathrm{AUTHIRE}_{is} + \gamma\mathrm{SCHOOL}_{is} + a_s + \varepsilon_{is} \quad (3)$$

$$\mathrm{OFT}_{is} = \delta + \alpha\mathrm{LACKPERS}_{is} + \gamma\mathrm{SCHOOL}_{is} + a_s + \varepsilon_{is} \quad (4)$$

$$\mathrm{OFT}_{is} = \delta + \alpha_1\mathrm{ADMINL}_{is} + \alpha_2\mathrm{AUTHIRE}_{is} + \alpha_3\mathrm{LACKPERS}_{is} + \gamma\mathrm{SCHOOL}_{is} + a_s + \varepsilon_{is}. \quad (5)$$

Equations 2–5 quantify a valid relationship if the assumption holds that the relationship between the

Table 2 Description of school-level control variables

Variables	Coding
School location	Rural area: population fewer than 3,000
	Small town: population from 3,000 to about 15,000
	Town: population from 15,000 to 100,000
	City: population from 100,000 to about 100,000,000
	Large city: population more than 1,000,000
School size	Small: fewer than 300 students
	Medium: 300–1,000 students
	Large: more than 1,000 students
Parental education	Low: percentage of parents who have ISCED 5 education is below national median and percentage of parents who have ISCED 3 education is below national median
	Medium: (1) percentage of parents who have ISCED 5 education is below national median and percentage of parents who have ISCED 3 education is above national median; (2) percentage of parents who have ISCED 5 education is above national median and percentage of parents who have ISCED 3 education is below national median
	High: percentage of parents who have ISCED 5 education is above national median and percentage of parents who have ISCED 3 education is above national median
Female	Percentage of female teachers
Part-time	Percentage of teachers who work part-time
Contract length	Percentage of teachers with short, fixed-term contracts
Educational attainment	Percentage of teachers with a master's degree or above
Teaching experience 5 years	Percentage of teachers with <5 years of experience
Teaching experience 20 years	Percentage of teachers with 20 or more years of experience

Table 3 Breakdown of variability in school-level out-of-field teaching in math and science

Country	Proportion of variability explained among schools within a region	Proportion of variability explained among regions	Gini coefficient[a]	2008 GDP per capita (PPP)[b]
Brazil	92.96	7.04	53.9	10,524.65
Bulgaria	97.86	2.14	45.3	13,187.39
Estonia	94.72	5.28	30	20,319.87
Hungary	96.54	3.46	24.7	19,460.01
Korea	95.74	4.26	31	27,707.05
Lithuania	98.22	1.78	37.6	19,138.18
Malta	100.00	0.00	26	24,768.97
Mexico	94.19	5.81	51.7	14,506.46
Norway	94.04	5.96	25	52,839.78
Portugal	95.70	4.30	38.5	23,079.82
Slovak Rep	88.23	11.77	26	21,995.00
Slovenia	94.60	5.40	28.4	29,605.53
Spain	97.61	2.39	32	30,847.93
Turkey	87.68	12.32	40.2	13,107.55
Belgium (F)	87.18	12.82	28	36,249.42

Region is defined by school location. GDP is measured in current dollars

[a] CIA World Factbook (n.d.)

[b] International Monetary Fund (2012)

variables of interest and the school-level percentage of out-of-field teaching is linear. However, because the outcome variable is a percentage measure that is bounded by zero and one, the linear functional form of these models may not be appropriate, especially at the boundaries. This specification issue suggests that the effect of the predictors may not be constant throughout the range of predictors. Although it is possible to include a non-linear form of predictors to augment Eqs. 2–5, there is no guarantee that the predicted values from the OLS specifications will lie within the unit interval. As an alternative, I fitted the same data to a set of fractional response models (FRM) introduced by Papke and Wooldridge (1996).[6] I specify four non-linear models in Eqs. 6–9:

$$\log\mathrm{it}(\mathrm{OFT}_{is}) = \delta + \alpha\mathrm{ADMINL}_{is} + \gamma\mathrm{SCHOOL}_{is} + a_s + \varepsilon_{is}, \tag{6}$$

$$\log\mathrm{it}(\mathrm{OFT}_{is}) = \delta + \alpha\mathrm{AUTHIRE}_{is} + \gamma\mathrm{SCHOOL}_{is} + a_s + \varepsilon_{is}, \tag{7}$$

$$\log\mathrm{it}(\mathrm{OFT}_{is}) = \delta + \alpha\mathrm{LACKPERS}_{is} + \gamma\mathrm{SCHOOL}_{is} + a_s + \varepsilon_{is}, \tag{8}$$

$$\log\mathrm{it}(\mathrm{OFT}_{is}) = \delta + \alpha_1\mathrm{ADMINL}_{is} + \alpha_2\mathrm{AUTHIRE}_{is} + \alpha_3\mathrm{LACKPERS}_{is} + \gamma\mathrm{SCHOOL}_{is} + a_s + \varepsilon_{is}. \tag{9}$$

The predictors and control variables are identical to those used in Eqs. 2–5. In addition, because administrative leadership may be affected by how much autonomy a school has, especially in the cross-national context, I also test the interaction term between these two variables in the full specification.

$$\mathrm{OFT}_{is} = \delta + \alpha_1\mathrm{ADMINL}_{is} + \alpha_2\mathrm{AUTHIRE}_{is} + \alpha_3\mathrm{ADMINL}_{is} * \mathrm{AUTHIRE}_{is} + \alpha_4\mathrm{LACKPERS}_{is} + \gamma\mathrm{SCHOOL}_{is} + a_s + \varepsilon_{is}, \tag{10}$$

$$\log\mathrm{it}(\mathrm{OFT}_{is}) = \delta + \alpha_1\mathrm{ADMINL}_{is} + \alpha_2\mathrm{AUTHIRE}_{is} + \alpha_3\mathrm{ADMINL}_{is} * \mathrm{AUTHIRE}_{is} + \alpha_4\mathrm{LACKPERS}_{is} + \gamma\mathrm{SCHOOL}_{is} + a_s + \varepsilon_{is}. \tag{11}$$

All regression analysis used proper school weights. TALIS employed a complex survey design with multiple stages of clustering and stratification. In order to compensate the design to reflect the true characteristics of the intended population, OECD uses a technique called balanced repeated replication (BRR) in the post hoc estimation of sampling variability of a statistic (OECD 2010). This same technique has been used in other OECD data set such as Programme for International Student Assessment (PISA). Most statistical package can directly deal with BRR weights in the estimation process (cite). It has been advised by experts designing these large-scale data project to use replication weights in the modeling process (Rutkowski et al. 2010). Under multilevel setting, major contemporary statistical packages can handle complex survey design weights (Carle 2009; Meinck and Vandenplas 2012; Rabe-Hesketh and Skrondal 2008). For the purpose of this study, I use Stata to fit series of fixed-effects models. Under the convention of fixed-effects studies (Chudgar et al. 2013; Hanushek and Woessmann 2010), I directly apply statistical weights in the estimation procedure to take account of data clustering without invoking BRR procedure.

Results

Current Status of Out-of-Field Teaching

In Fig. 1, I first show the percentage of OFTs in math and science by country. The extent of out-of-field teaching ranged from 1.97 % in Hungary to 15.7 % in Brazil. The international mean is 10.3 %, which means one in ten teachers in math and science was teaching out-of-field. Compared to statistics from the United States, TALIS countries display a low level of out-of-field teaching.[7] While the country-level averages indicate cross-country variation in out-of-field teaching, they do not reveal variation within countries. In Table 3, I partition the variation in school percentage of out-of-field teaching into two sources in each country: among regions and among schools within a region.[8] The estimates indicate that there is greater variation in the level of out-of-field teaching among schools in the same region than between schools in two different regions. This pattern holds across all TALIS countries, regardless of each country's level of economic development or distribution of income, as measured by

[6] Note that, the estimates from FRM models are not directly comparable to those from OLS. I calculated the partial effect of each estimate as suggested by Wooldridge (2010). A fully robust sandwich estimator is implemented using the QMLE estimation method.

[7] Readers should bear in mind that the estimates from TALIS are not directly comparable to those from the United States because of different measurements of OFT. TALIS employs a more general definition. Because of the wording of the survey item, individual respondents may interpret the out-of-field question differently. For instance, someone with high school–level academic training could justify this training as sufficient. By using this measure, the survey probably underestimates out-of-field teaching levels because of its relaxed definition of academic training. The true levels of out-of-field teaching may be higher, as suggested by estimates from other international databases such as TIMSS.

[8] I partition the variance by specifying a basic two-level fully unconditional linear model within each country.

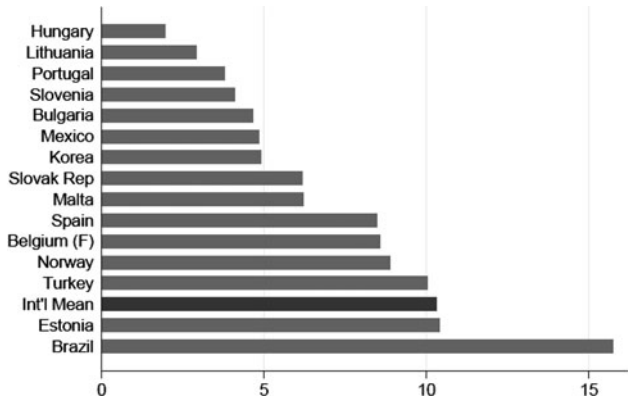

Fig. 1 National average of OFTs in mathematics and science at lower secondary level (as percentage)

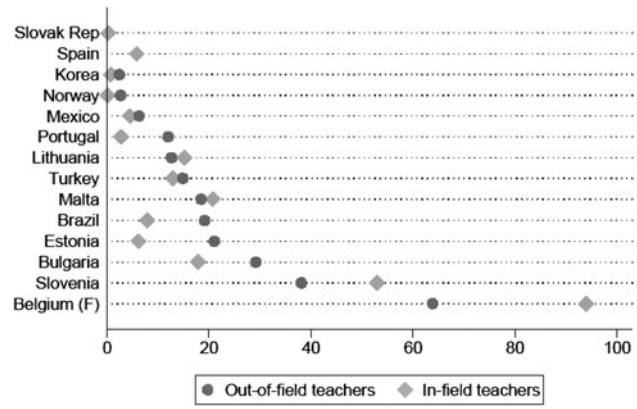

Fig. 2 Percentage of out-of-field and in-field teachers who do not have a bachelor's degree by country

GPD per capita and the Gini coefficient, respectively. It is easy to understand why this happens in more developed and more equal nations, since regional development is more likely to be equalized, which translates into equal distribution of school resources and teachers. But it is interesting to observe that even in developing nations such as Brazil and Turkey, variability among schools exceeds variability among regions. This finding confirms the school perspective that inequalities in teacher staffing are associated with certain types of schools. The deployment of OFTs in math and science fields happens in these schools.

In Fig. 2, I show the percentage of teachers who do not have a bachelor's degree by out-of-field and in-field status.[9] It is clear that in the majority of cases, the two percentages are fairly close. This suggests that a considerable number of OFTs do at least have a bachelor's degree. Out-of-field teaching does not seem to be an issue of qualifications.

Correlation of School Characteristics and Levels of Out-of-Field Teaching in Math and Science

The results are summarized in Tables 4 and 5. In Table 4, columns 2 to 5 present estimates from the first set of variables. Columns 2 and 3 show the estimates obtained without using country fixed effects, and columns 4 and 5 show estimates obtained by controlling for country fixed effects. The OLS columns contain estimates obtained through OLS estimation, and the FRM columns contain estimates obtained through FRM. Columns 6 to 13 repeat the same strategy with other sets of variables. The structure of Table 5 is similar. Columns 2 to 5 present estimates from the specification in which all three main predictors are included. Estimates shown in columns 6 to 9 are obtained by adding interaction terms.

First, the estimates from OLS and FRM models are very close. This suggests that the functional form may not be a significant issue in estimating a school's reliance on out-of-field assignments. Second, neither the administrative leadership nor the personnel shortage variables show significance in any specifications. This is rather surprising, because it suggests that the principal's leadership style and the teacher supply do not correlate with a school's decision to deploy OFTs. The only variable that allows significant interpretation is the autonomy-of-hiring variable. The estimates suggest that if schools are given one standard deviation more autonomy in decision-making, the percentage of out-of-field teaching will decline by 2.8–3.4 %. Considering that cross-nationally the international average of out-of-field teaching in math and science is 10.3 %, such a change would translate to about 30 percent fewer OFTs in these subject fields.

When country fixed effects are accounted for, the estimates become insignificant in both OLS and FRM specifications. This suggests that once the country specifics are controlled for, school autonomy in hiring teachers and determining salaries does not affect a school's decision to deploy OFTs in math and science. This is not surprising because the variation in autonomy is limited within each country, therefore reducing the association obtained cross-nationally.

School Autonomy and Reliance on OFTs

The autonomy-of-hiring variable is constructed based on five elements: the responsibility of selecting teachers for hiring; the responsibility of firing teachers; the responsibility of establishing teachers' starting salaries; the responsibility of determining teachers' salary increases; and the responsibility of allocating funds for teachers'

[9] Hungary is excluded because all of the Hungarian teachers in the TALIS sample held a bachelor's degree.

Table 4 Pooled OLS and FRM estimates of relationships between school characteristics and out-of-field teaching in mathematics and science, including country fixed effects

Models	Equations 2 and 6				Equations 3 and 7				Equations 4 and 8			
Variables	OLS	FRM	OLS	FRM	OLS	FRM	OLS	FRM	OLS	FRM	OLS	FRM
Administrative leadership	0.008	0.007	0.010	0.007								
	(0.013)	(0.014)	(0.015)	(0.014)								
Autonomy of hiring					−0.028*	−0.028^{+}	0.004	−0.028^{+}				
					(0.014)	(0.016)	(0.030)	(0.017)				
Lack of school personnel									−0.003	−0.005	−0.012	−0.005
									(0.014)	(0.014)	(0.016)	(0.015)
School size[a]												
Medium	−0.007	0.001	−0.016	0.001	−0.019	−0.010	−0.025	−0.010	−0.013	−0.005	−0.024	−0.005
	(0.036)	(0.036)	(0.038)	(0.034)	(0.039)	(0.038)	(0.040)	(0.036)	(0.037)	(0.036)	(0.039)	(0.035)
Large	−0.037	−0.030	−0.049	−0.030	−0.057	−0.048	−0.062	−0.048	−0.043	−0.035	−0.056	−0.035
	(0.037)	(0.036)	(0.044)	(0.039)	(0.041)	(0.040)	(0.046)	(0.041)	(0.038)	(0.037)	(0.046)	(0.040)
School location[b]												
Small town	−0.098**	−0.092**	−0.099**	−0.092*	−0.106**	−0.096**	−0.110**	−0.096**	−0.098**	−0.090*	−0.098*	−0.090*
	(0.034)	(0.035)	(0.038)	(0.036)	(0.036)	(0.036)	(0.040)	(0.037)	(0.035)	(0.036)	(0.039)	(0.037)
Town	−0.063	−0.059	−0.066	−0.059	−0.065	−0.056	−0.073	−0.056	−0.065	−0.059	−0.067	−0.059
	(0.040)	(0.044)	(0.046)	(0.044)	(0.042)	(0.046)	(0.048)	(0.045)	(0.042)	(0.044)	(0.047)	(0.045)
City	−0.071^{+}	−0.068	−0.069	−0.068	−0.076^{+}	−0.069	−0.076	−0.069	−0.067	−0.061	−0.057	−0.061
	(0.041)	(0.042)	(0.048)	(0.046)	(0.044)	(0.044)	(0.051)	(0.047)	(0.042)	(0.042)	(0.049)	(0.047)
Large city	−0.098**	−0.102**	−0.107**	−0.102**	−0.103**	−0.101**	−0.109**	−0.101*	−0.098**	−0.100**	−0.104*	−0.100*
	(0.034)	(0.036)	(0.040)	(0.039)	(0.036)	(0.038)	(0.042)	(0.040)	(0.035)	(0.038)	(0.041)	(0.039)
Parental education[c]												
Medium	0.049	−0.043	0.056^{+}	−0.043	0.055	−0.049	0.061^{+}	−0.049	0.053	−0.046	0.061^{+}	−0.046
	(0.033)	(0.032)	(0.034)	(0.030)	(0.035)	(0.033)	(0.035)	(0.031)	(0.033)	(0.032)	(0.035)	(0.031)
High	0.035	−0.011	0.031	−0.011	0.038	−0.015	0.034	−0.015	0.033	−0.017	0.029	−0.017
	(0.029)	(0.030)	(0.031)	(0.027)	(0.031)	(0.032)	(0.032)	(0.028)	(0.029)	(0.031)	(0.032)	(0.028)
Percentage of female teachers	0.000	0.000	0.000	0.000	0.000	0.000	−0.000	0.000	0.000	0.000	−0.000	0.000
	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)
Percentage of teachers who work part-time	−0.001	−0.001	−0.002*	−0.001^{+}	−0.001*	−0.001*	−0.002**	−0.001*	−0.001^{+}	−0.001^{+}	−0.002**	−0.001^{+}
	(0.001)	(0.001)	(0.001)	(0.000)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.000)
Percentage of teachers on short, fixed-term contracts	0.002^{+}	0.001^{+}	0.002^{+}	0.001*	0.001	0.001	0.001^{+}	0.001^{+}	0.002^{+}	0.001^{+}	0.002^{+}	0.001*
	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)
Percentage of teachers with a master's degree or above	−0.001**	−0.001*	0.001	−0.001*	−0.001**	−0.001**	0.001	−0.001**	−0.001**	−0.001*	0.001	−0.001*

Table 4 continued

Models	Equations 2 and 6				Equations 3 and 7				Equations 4 and 8			
Variables	OLS	FRM	OLS	FRM	OLS	FRM	OLS	FRM	OLS	FRM	OLS	FRM
	(0.000)	(0.001)	(0.001)	(0.001)	(0.000)	(0.001)	(0.001)	(0.001)	(0.000)	(0.001)	(0.001)	(0.001)
Percentage of teachers with <5 years of experience	0.000	−0.000	0.000	−0.000	0.000	0.000	0.000	0.000	−0.000	−0.000	−0.000	−0.000
	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)
Percentage of teachers with 20 or more years of experience	−0.000	−0.001	0.000	−0.001	−0.000	−0.001	0.000	−0.001	−0.001	−0.001	−0.000	−0.001
	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)
Country fixed effects	No	No	Yes	Yes	No	No	Yes	Yes	No	No	Yes	Yes
N	2083	2083	2083	2083	1986	1986	1986	1986	2012	2012	2012	2012

Standard errors in parentheses

[a] Small schools as reference group

[b] Rural schools as reference group

[c] Low school-level parental education as reference group

$^{+}$ $p < 0.10$; * $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$

professional development. For each of these aspects, principals were asked to indicate who, from a list of actors, held greater responsibility: the principal, teachers, the school governing board, the regional or local educational authority, or the national educational authority. Therefore, each element consists of a series of zero/one binary choices. For instance, to identify who is responsible for selecting and hiring teachers, the principals responded to the following:

- The principal is more responsible (yes/no).
- Teachers are more responsible (yes/no).
- The school governing board is more responsible (yes/no).
- The regional or local educational authority is more responsible (yes/no).
- The national education authority is more responsible (yes/no).

This information provided an opportunity to look into the school autonomy constructs and to identify specific ingredients of institutional arrangements that could potentially affect OFT deployment. Of course, principals could identify multiple elements of school autonomy simultaneously. To simplify this exercise, I focused on each of the five elements individually to test their relationship with school-level out-of-field teaching. I replaced the autonomy-of-hiring variable with these binary predictors and ran a series of models similar to those specified in Eqs. 5 and 9. The findings are presented in Table 6.[10] The presentation of Table 6 is similar to those of Tables 4 and 5 except that estimates are all obtained using fixed-effects models.

Although these measures are crude in themselves, they give us a basic sense of the importance of particular elements in school management that can relate to out-of-field teaching assignments. School-level out-of-field teaching is considerably higher when either teachers or the regional educational authority are given more responsibility to decide on teachers' starting salaries. In schools where the increase in teacher salaries is determined by a regional educational authority, levels of out-of-field teaching are also higher. By contrast, when principals are given more responsibility to determine the salary increases of teachers, levels of out-of-field teaching are somewhat lower. All these findings are robust after controlling for country fixed effects. These findings point to the importance of delegating staffing powers to local schools. In contrast to centralizing authority, which tends to correlate with more out-of-field teaching, giving local schools more power seems to effectively reduce the hiring of OFTs.

[10] I show only results with statistical significance. Specifications in which no statistical significance was detected are not shown.

Table 5 Pooled OLS and FRM estimates of relationships between school characteristics and out-of-field teaching in mathematics and science, including country fixed effects in alternative specifications

Model	Equations 5 and 9				Equations 10 and 11			
Variables	OLS	FRM	OLS	FRM	OLS	FRM	OLS	FRM
Administrative leadership	0.009	0.007	0.007	0.006	0.007	0.008	0.011	0.012
	(0.014)	(0.015)	(0.016)	(0.015)	(0.020)	(0.025)	(0.021)	(0.027)
Autonomy of hiring	−0.032*	−0.034*	−0.000	−0.005	−0.031*	−0.034*	−0.002	−0.008
	(0.014)	(0.017)	(0.030)	(0.029)	(0.013)	(0.017)	(0.029)	(0.029)
Leadership autonomy					−0.002	0.001	0.004	0.005
					(0.016)	(0.021)	(0.016)	(0.021)
Lack of school personnel	−0.008	−0.011	−0.013	−0.015	−0.008	−0.011	−0.013	−0.015
	(0.015)	(0.015)	(0.017)	(0.015)	(0.015)	(0.015)	(0.017)	(0.015)
School size[a]								
Medium	−0.026	−0.021	−0.034	−0.028	−0.026	−0.020	−0.034	−0.028
	(0.039)	(0.039)	(0.041)	(0.039)	(0.039)	(0.039)	(0.041)	(0.039)
Large	−0.061	−0.055	−0.067	−0.059	−0.061	−0.055	−0.067	−0.058
	(0.041)	(0.040)	(0.049)	(0.046)	(0.041)	(0.041)	(0.049)	(0.046)
School location[b]								
Small town	−0.102**	−0.088*	−0.104*	−0.087*	−0.102**	−0.088*	−0.104*	−0.087*
	(0.037)	(0.037)	(0.041)	(0.040)	(0.038)	(0.037)	(0.041)	(0.040)
Town	−0.069	−0.055	−0.075	−0.059	−0.068	−0.055	−0.075	−0.059
	(0.043)	(0.046)	(0.049)	(0.049)	(0.044)	(0.047)	(0.049)	(0.049)
City	−0.074^{+}	−0.061	−0.067	−0.051	−0.074^{+}	−0.061	−0.067	−0.050
	(0.043)	(0.043)	(0.051)	(0.055)	(0.043)	(0.043)	(0.051)	(0.055)
Large city	−0.102**	−0.098*	−0.107*	−0.104**	−0.102**	−0.098*	−0.107*	−0.104**
	(0.037)	(0.038)	(0.043)	(0.040)	(0.037)	(0.038)	(0.043)	(0.040)
Parental education[c]								
Medium	0.063^{+}	−0.056	0.071*	−0.062^{+}	0.063^{+}	−0.056	0.071*	−0.062^{+}
	(0.035)	(0.034)	(0.036)	(0.033)	(0.035)	(0.034)	(0.036)	(0.033)
High	0.037	−0.024	0.034	−0.037	0.037	−0.024	0.034	−0.037
	(0.031)	(0.033)	(0.033)	(0.030)	(0.031)	(0.033)	(0.033)	(0.030)
Percentage of female teachers	0.000	0.000	−0.000	−0.000	0.000	0.000	−0.000	−0.000
	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)
Percentage of teachers who work part-time	−0.001*	−0.001*	−0.002**	−0.002**	−0.001*	−0.001*	−0.002**	−0.002**
	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)
Percentage of teachers on short, fixed-term contracts	0.001	0.001	0.001^{+}	0.001^{+}	0.001	0.001	0.001^{+}	0.001^{+}
	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)
Percentage of teachers with a master's degree or above	−0.001*	−0.001*	0.001	0.001	−0.001*	−0.001*	0.001	0.001
	(0.000)	(0.001)	(0.001)	(0.001)	(0.000)	(0.001)	(0.001)	(0.001)
Percentage of teachers with <5 years of experience	−0.000	−0.000	−0.000	−0.000	−0.000	−0.000	−0.000	−0.000
	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)
Percentage of teachers with 20 or more years of experience	−0.001	−0.001	−0.000	−0.000	−0.001	−0.001	−0.000	−0.000
	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)
Country fixed effects	No	No	Yes	Yes	No	No	Yes	Yes
N	1911	1911	1911	1911	1911	1911	1911	1911

Standard errors in parentheses

[a] Small schools as reference group

[b] Rural schools as reference group

[c] Low school-level parental education as reference group

$^{+}$ $p < 0.10$; * $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$

Table 6 Pooled OLS and FRM estimates of relationships between school autonomy constructs and out-of-field teaching in mathematics and science, including country fixed effects

Variables	OLS	FRM	OLS	FRM	OLS	FRM	OLS	FRM
Administrative leadership	0.012	0.011	0.012	0.012	0.012	0.010	0.013	0.013
	(0.015)	(0.015)	(0.015)	(0.015)	(0.015)	(0.015)	(0.015)	(0.014)
Teachers are more responsible for establishing their starting salaries	0.127	0.178^{+}						
	(0.106)	(0.091)						
Regional educational authority is more responsible for establishing teachers' starting salaries			0.091**	0.147**				
			(0.034)	(0.047)				
Principal is more responsible for determining teachers' salary increases					−0.060^{+}	−0.112*		
					(0.034)	(0.054)		
Regional educational authority is more responsible for determining teachers' salary increases							0.100**	0.148***
							(0.036)	(0.045)
Lack of school personnel	−0.013	−0.015	−0.013	−0.016	−0.013	−0.016	−0.012	−0.014
	(0.017)	(0.015)	(0.017)	(0.015)	(0.017)	(0.015)	(0.016)	(0.014)
School size[a]								
Medium	−0.024	−0.017	−0.024	−0.017	−0.026	−0.021	−0.026	−0.017
	(0.040)	(0.038)	(0.039)	(0.038)	(0.039)	(0.038)	(0.039)	(0.038)
Large	−0.054	−0.046	−0.052	−0.044	−0.057	−0.052	−0.051	−0.042
	(0.047)	(0.044)	(0.046)	(0.044)	(0.046)	(0.044)	(0.046)	(0.045)
Parental education[b]								
Small town	−0.096*	−0.083*	−0.092*	−0.076*	−0.095*	−0.078*	−0.092*	−0.078*
	(0.039)	(0.039)	(0.039)	(0.038)	(0.039)	(0.037)	(0.039)	(0.039)
Town	−0.068	−0.056	−0.063	−0.044	−0.063	−0.044	−0.067	−0.054
	(0.047)	(0.047)	(0.046)	(0.045)	(0.047)	(0.046)	(0.046)	(0.045)
City	−0.059	−0.045	−0.057	−0.038	−0.055	−0.035	−0.058	−0.041
	(0.049)	(0.052)	(0.048)	(0.051)	(0.048)	(0.051)	(0.048)	(0.052)
Large city	−0.103*	−0.102**	−0.097*	−0.094*	−0.102*	−0.095*	−0.101*	−0.099*
	(0.041)	(0.039)	(0.041)	(0.038)	(0.041)	(0.037)	(0.040)	(0.038)
School SES[c]								
Medium	0.064^{+}	−0.056^{+}	0.067^{+}	−0.062^{+}	0.066^{+}	−0.059^{+}	0.068*	−0.065*
	(0.035)	(0.033)	(0.035)	(0.033)	(0.035)	(0.032)	(0.035)	(0.032)
High	0.028	−0.036	0.028	−0.044	0.030	−0.038	0.030	−0.044
	(0.032)	(0.029)	(0.032)	(0.029)	(0.032)	(0.029)	(0.032)	(0.029)
Percentage of female teachers	−0.000	−0.000	−0.000	−0.000	−0.000	−0.000	−0.000	−0.000
	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)
Percentage of teachers who work part-time	−0.002**	−0.002**	−0.002**	−0.002**	−0.002**	−0.002**	−0.002**	−0.002**
	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)

Table 6 continued

Variables	OLS	FRM	OLS	FRM	OLS	FRM	OLS	FRM
Percentage of teachers on short, fixed-term contracts	0.002^{+}	0.001*	0.002^{+}	0.001*	0.002^{+}	0.001*	0.002^{+}	0.001*
	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)
Percentage of teachers with a master's degree or above	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001
	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)
Percentage of teachers with <5 years of experience	−0.000	−0.000	0.000	−0.000	0.000	−0.000	−0.000	−0.000
	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)
Percentage of teachers with 20 or more years of experience	−0.000	−0.000	−0.000	−0.000	−0.000	−0.000	−0.000	−0.000
	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)
Country fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
N	1983	1983	1983	1983	1978	1978	1977	1977

Standard errors in parentheses

[a] Small schools as reference group

[b] Rural schools as reference group

[c] Low school-level parental education as reference group

$^{+}$ $p < 0.10$; * $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$

Several school characteristics are associated with prevalence of out-of-field teaching in Tables 4 and 5. Rural schools are consistently more likely than schools in other regions to have a large number of OFTs in math and science. The difference ranges from 9 to 10 % depending on specification. On the other hand, schools in large cities have a much lower percentage of OFTs. Interestingly, schools that hire more part-time teachers also have lower levels of out-of-field teaching. Schools that have more teachers on flexible contracts (i.e., without a permanent position) use more OFTs. Schools whose teaching force contains a higher proportion of teachers with a master's degree or above have lower levels of out-of-field teaching. These effects are very small in magnitude, though they are statistically significant.

Discussion

Readers should be first reminded that the TALIS sample is not a representative sample of every nation in the world. In particular, there are fewer developing countries. Therefore, the results only speak to these participating nations but not beyond.

This exploratory study is based on cross-sectional data. Although the fixed-effects approach allows me to overcome the endogeneity issue at the country level, the findings are nonetheless associational rather than causal. My analysis has three implications. First, I revisited the debate on the correlates of out-of-field teaching from an international comparative perspective. I showed that out-of field teaching is not a teacher qualification issue, because the majority of OFTs do have a bachelor's degree—they simply lack training in the subject they are teaching. The deployment of OFTs in math and science does not relate to teacher shortages. In fact, the teacher shortage variable is not significant in any of the specifications. I found most evidence to support the school organizational perspective in explaining variation in levels of out-of-field teaching across countries. In each country, regardless of the level of economic development and income equality, most of the variability in out-of-field teaching occurs among schools. Regression models confirm that several school characteristics are significantly associated with school-level out-of-field teaching in math and science. Across nations, rural schools, schools with more teachers on short, fixed-term contracts, and schools with fewer teachers with a master's degree are more likely to rely on OFTs. Schools in large cities have significantly fewer OFTs. Schools with more part-time teachers also use fewer OFTs, though the difference is trivial. These findings are consistent with previous literature on the distribution of teachers (Lankford et al. 2002; Luschei and Carnoy 2009). They indicate that the distribution of OFTs is associated with the structure of

the teacher labor market, in which schools with certain characteristics are in a disadvantaged position to recruit high-quality teachers.

Second, although out-of-field teaching seems to be a school-specific issue, international evidence suggests that a principal's administrative leadership style does not play a major part in causing variations in out-of-field teaching. In contrast to the limited role of a principal's administrative leadership, I found that giving schools more autonomy has a greater impact on out-of-field teaching. However, this effect is dependent on the country context: once the country specifics are accounted for, the significance of the autonomy variable vanishes.

Third, I broke the autonomy variable down into five aspects of school management and tested the relationship of each of the five aspects to school-level out-of-field teaching. I found that giving the principal more authority to determine teachers' salary increases within a school is associated with lower levels of out-of-field teaching in the school, compared to a situation in which the authority to determine salary increases is held at the level of the regional educational authority.

In view of these results, it seems contradictory to claim that a principal's administrative leadership does not matter, but at the same time that giving the principal more authority in determining teacher salaries could change the deployment of OFTs. I argue that the extent to which a principal can exert influence in a school is constrained by the power delegated to him or her by the institutional arrangement. This is not to deny that the principal's agency could overcome the constraints facing him or her. Yet in the statistical sense, for an average school, the institutional arrangement comes first by defining the perimeter within which the administrators can operate. This echoes previous findings that the locus of decision-making is key to determining important features of teacher labor markets (Luschei and Carnoy 2009). In addition to the principal's autonomy in deciding teachers' salary increases, school characteristics such as location and the percentage of teachers with master's degrees are all part of the underlying administrative regimes that are shaped by policies. While this study does not focus on prescribing institutional arrangements that can lead to lower levels of out-of-field teaching, international evidence suggests that certain arrangements could be leveraged to achieve higher equity and efficiency (Carnoy et al. 2007; Luschei et al. forthcoming).

Finally, the question arises why the ability to determine salary increases could change the way in which out-of-field assignments are made. Though there are no proven theories to explain this finding, there are several possibilities worthy of future analysis. One is that allowing each principal to determine the salary structure in his or her school essentially creates a more dynamic labor market that allows schools to use wages to signal their demand for skills in particular subject fields. The incentive structure will thus reflect what type of teacher is really needed in the classroom. This permits more efficient distribution of teachers' subject matter skills both across schools and within schools. It is also possible that the delegation of more responsibility to the school principal implicitly requires the principal to monitor teachers' skills closely and consequently to make better assignment decisions. Because of the lack of detailed data on school staffing and teacher compensation, it is difficult to test this hypothesis explicitly. Future research is needed to explore this issue.

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References

Akiba, M., & LeTendre, G. K. (2009). *Improving teacher quality: The US teaching force in global context*. New York, London: Teachers College Press.

Akiba, M., LeTendre, G. K., & Scribner, J. P. (2007). Teacher quality, opportunity gap, and national achievement in 46 countries. *Educational Researcher, 36*(7), 369–387.

Ammermueller, A., & Pischke, J.-S. (2009). Peer effects in European primary schools: Evidence from the progress in International Reading Literacy Study. *Journal of Labor Economics, 27*(3), 315–338. doi:10.1086/603650.

Carle, A. C. (2009). Fitting multilevel models in complex survey data with design weights: Recommendations. *BMC Medical Research Methodology, 9*, 49. doi:10.1186/1471-2288-9-49.

Carnoy, M., Gove, A. K., & Marshall, J. H. (2007). *Cuba's academic advantage: Why students in Cuba do better in school*. Stanford, CA: Stanford University Press.

Central Intelligence Agency. (n.d.). The World Factbook. Retrieved from https://www.cia.gov/library/publications/the-world-factbook/. Accessed May 2012.

Chudgar, A., Luschei, T. F., & Zhou, Y. (2013). Science and mathematics achievement and the importance of classroom composition: Multicountry analysis using TIMSS 2007. *American Journal of Education, 119*(2), 295–316.

Eurydice (The Information Network on Education in Europe). (2002). The teaching profession in Europe (profile, trends and concerns) report 2, supply and demand, general lower secondary education.

Hanushek, E. A., & Woessmann, L. (2010). The economics of international differences in educational achievement. In E. A. Hanushek, S. Machin, & L. Woessmann (Eds.), *Handbook of the economics of education* (Vol. 3, pp. 89–200). Amsterdam: North-Holland.

Hechinger, F. M. (1985, October 8). About education; "dirty, little secret" of unlicensed teachers. *The New York Times*. Retrieved from http://www.nytimes.com/1985/10/08/science/about-education-dirty-little-secret-of-unlicensed-teachers.html?scp=23&sq=Albert+Shanker&st=nyt. Accessed May 2012.

Ingersoll, R. M. (1999). The problem of underqualified teachers in American secondary schools. *Educational Researcher, 28*(2), 26–37.

Ingersoll, R. M. (2002). *Out-of-field teaching, educational inequality, and other organization of schools: An exploratory analysis*

(Document R-02-1). Seattle, WA: Center for the Study of Teaching and Policy.

Ingersoll, R. M. (2003). *Out-of-field teaching and the limits of teacher policy (Document R-03-5)*. Seattle, WA: Center for the Study of Teaching and Policy.

Ingersoll, R. M. (2005). The problem of underqualified teachers: A sociological perspective. *Sociology of Education, 78*(2), 175–178.

Ingersoll, R. M. (2006). *Who controls teachers' work? Power and accountability in America's schools*. Cambridge, MA: Harvard University Press.

Ingersoll, R. M., & Perda, D. (2010). Is the supply of mathematics and science teachers sufficient? *American Educational Research Journal, 47*(3), 563–594. doi:10.3102/0002831210370711.

International Monetary Fund. (2012). World Economic Outlook Database, October 2012. Retrieved from http://www.imf.org/external/pubs/ft/weo/2012/02/weodata/index.aspx. Accessed Oct 2012.

Ladd, H. F. (2007). Teacher labor markets in developed countries. *The Future of Children, 17*(1), 201–217.

Lankford, H., Loeb, S., & Wykoff, J. (2002). Teacher sorting and the plight of urban schools: A descriptive analysis. *Educational Evaluation and Policy Analysis, 24*(1), 37–62. doi:10.3102/01623737024001037.

Luschei, T. F., & Carnoy, M. (2009). Educational production and the distribution of teachers in Uruguay. *International Journal of Educational Development, 30*(2), 169–181.

Luschei, T. F., Chudgar, A., & Rew, W. (2013). Exploring differences in the distribution of teacher qualifications across Mexico and South Korea: Evidence from the Teaching and Learning International Survey. *Teachers College Record, 115*(5). http://www.tcrecord.org/Content.asp?ContentId=16964.

Meinck, S., & Vandenplas, C. (2012). Sample size requirement in HLM: An empirical study (IERI Monograph Series). Hamburg: IERI Institute

Morton, B. A., Peltola, P., Hurwitz, M. D., Orlofsky, G. F., Strizek, G. A., & Gruber, K. J. (2008). *Education and certification qualifications of departmentalized public high school-level teachers of core subjects: Evidence from the 2003–04 Schools and Staffing Survey*. Washington, DC: National Center for Education Statistics.

National Commission on Teaching and America's Future. (1996). *What matters most: Teaching for America's future*. New York, NY: National Commission on Teaching and America's Future.

Ohinata, A., & van Ours, J. C. (2011). *How immigrant children affect the academic achievement of native Dutch children* (Discussion Paper No. 6212). Bonn, Germany: Institute for the Study of Labor.

Organisation for Economic Co-operation and Development (OECD). (2005). *Teachers matter: Attracting, developing and retaining effective teachers*. Paris: OECD. doi:10.1787/9789264018044-en.

Organisation for Economic Co-operation and Development (OECD). (2009). *Creating effective teaching and learning environments: First results from TALIS*. Paris: OECD. doi:10.1787/9789264072992-en.

Organisation for Economic Co-operation and Development (OECD). (2010). *TALIS 2008 technical report*. Paris: OECD. doi:10.1787/9789264079861-en.

Papke, L. E., & Wooldridge, J. M. (1996). Econometric methods for fractional response variables with an application to 401(k) plan participation rates. *Journal of Applied Econometrics, 11*(6), 619–632.

Rabe-Hesketh, S., & Skrondal, A. (2008). *Multilevel and longitudinal modeling using Stata (2nd ed)* (p. 562). College Station, TX: Stata Press.

Rutkowski, L., Gonzalez, E., Joncas, M., & von Davier, M. (2010). International large-scale assessment data: Issues in secondary analysis and reporting. *Educational Researcher, 39*(2), 142–151. doi:10.3102/0013189X10363170.

Schools and Staffing Survey. (2010). Public school teacher data file, 2007–2008. Washington, DC: National Center for Education Statistics. Retrieved from http://nces.ed.gov/surveys/sass/tables/sass0708_006_t1n.asp. Accessed May 2012.

Seastrom, M. M., Gruber, K. J., Henke, R. R., McGrath, D. J., & Cohen, B. A. (2002). *Qualifications of the public school teacher workforce: Prevalence of out-of-field teaching*. Washington, DC: National Center for Education Statistics.

Shulman, L. S. (1986). Those who understand: Knowledge growth in teaching. *Educational Researcher, 15*(4), 4–14.

UNESCO Institute for Statistics. (2006). *Teachers and educational quality: Monitoring global needs for 2015*. Montreal: UNESCO Institute for Statistics.

Wooldridge, J. M. (2010). *Econometric analysis of cross section and panel data* (2nd ed.). Cambridge, MA: MIT Press.